

Job Portal Web Application

A PROJECT REPORT

Submitted by

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Project Overview

The Job Portal Web Application is a full-stack web-based system developed using **React.js**, **Node.js**, **Express.js**, and **MongoDB**. The main objective of this project is to provide a platform where job listings can be displayed dynamically and managed efficiently.

The system allows users to view available job opportunities fetched from a backend server connected to a database. The frontend provides an interactive user interface, while the backend handles API requests and database operations.

This project demonstrates how modern web applications are built using a client-server architecture and RESTful APIs.

Objectives

- To develop a dynamic job listing web application
 - To understand full-stack web development using MERN stack
 - To implement frontend-backend communication using REST APIs
 - To store and retrieve data using MongoDB database
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Technologies Used

Frontend:

- React.js
- HTML, CSS, JavaScript
- Axios (API calls)

Backend:

- Node.js
- Express.js

Database:

- MongoDB
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System Architecture

The system follows a **3-tier architecture**:

Frontend (React)



Backend (Node + Express API)



Database (MongoDB)

Flow of the Project

Step 1: User Interface (Frontend)

- User opens the website (React app)
 - Home page loads job listings UI
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Step 2: API Request

- React sends request using Axios

`axios.get("http://localhost:5000/api/jobs")`

Step 3: Backend Processing

- Node.js receives request
 - Express routes handle API logic
 - Data is fetched from MongoDB
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Step 4: Database Interaction

- MongoDB stores job details like:
 - Job title
 - Company
 - Salary
 - Description
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Step 5: Response to Frontend

- Backend sends data back in JSON format

- React displays job listings on UI
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Features of the Project

- Display of job listings dynamically
 - Backend API integration
 - MongoDB database storage
 - Responsive React-based UI
 - Modular architecture (frontend & backend separated)
 - Easy to extend with login/register system in future
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Modules in Project

1. Home Module

- Displays all available job listings

2. Backend API Module

- Handles requests like GET jobs, POST jobs

3. Database Module

- Stores job-related information
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Future Enhancements

- User authentication (Login/Register using JWT)
 - Admin panel for job posting
 - Job application system
 - Filtering and search functionality
 - Email notifications
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Conclusion

This project successfully demonstrates the development of a full-stack Job Portal application using the MERN stack. It helps in understanding frontend-backend integration, API handling, and database operations. The system can be further enhanced by adding authentication and advanced job management features.